Fixed-Mass Nodal Coercivity for Density-Weighted Curvature

An Independent Auxiliary Verification Note for WCT Computation

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Abstract

On a fixed two-dimensional periodic domain, we prove that positive conserved mass and an upper bound on the cubic–quintic projection-free Hamiltonian enforce a strictly positive and quantitatively explicit lower comparison between density-weighted Laplacian curvature and ordinary Laplacian energy. No uniform positive lower bound on pointwise amplitude is imposed, so fields with nodes are allowed. This closes a previously open *fixed-domain, fixed-resource* analytic step in a separate curvature–communication study. The constant is not controlled uniformly in domain size, spectral resolution, or computational input length. An exact two-site version and the algebraic readout-cost transfer are formalized in Lean. This auxiliary note is independent of the author's original P versus NP manuscript, which it neither revises nor claims to complete.

1 The maintained Hamiltonian and the curvature weight

Let $\Omega = \mathbb{R}^2/\Lambda$ be a fixed connected flat two-dimensional torus, of volume $V = |\Omega|$, and let $u \in H^2(\Omega;\mathbb{C})$. For $\kappa = k_\star^2 \geq 0$ and $\beta, \gamma > 0$, define

$$\mathcal{H}[u] = \int_\Omega \left(|(\Delta + \kappa)u|^2 - \frac{\beta}{2}|u|^4 + \frac{\gamma}{3}|u|^6 \right) dx. \tag{1}$$

This is the Hamiltonian of the separately maintained projection-free evolution

$$i\partial_t u = (\Delta + \kappa)^2 u - \beta|u|^2 u + \gamma|u|^4 u. \tag{2}$$

The present theorem concerns the *distinct* density-weighted curvature functional, not the Hamiltonian's own quadratic term. For $\varepsilon > 0$, $\alpha \geq 0$, let

$$F_{\varepsilon,\alpha}(s) = \frac{s}{(s + \varepsilon^2 e^{-2\alpha s})^2}, \qquad \mathcal{Q}_{\varepsilon,\alpha}[u] = \int_\Omega F_{\varepsilon,\alpha}(|u|^2)|\Delta u|^2 \, dx. \tag{3}$$

The weight vanishes at every node $u = 0$. Pointwise coercivity is therefore false without a node-free assumption, and the result below is genuinely global.

2 Conservation-derived compactness

Put $a_{\beta,\gamma} = 3\beta^2/(16\gamma)$. For every $s \geq 0$,

$$-\frac{\beta}{2}s^2 + \frac{\gamma}{3}s^3 + a_{\beta,\gamma}s = \frac{\gamma}{3}s\left(s - \frac{3\beta}{4\gamma}\right)^2 \geq 0. \tag{4}$$

Consequently

$$\|(\Delta + \kappa)u\|_2^2 \leq \mathcal{H}[u] + a_{\beta,\gamma}\,\|u\|_2^2\,. \tag{5}$$

Given a positive mass $m > 0$ and finite $H_\star \in \mathbb{R}$, define

$$\mathcal{A}_{m,H_\star} = \{u \in H^2(\Omega;\mathbb{C}) : \|u\|_2^2 = m, \quad \mathcal{H}[u] \le H_\star\}.$$

If this class is nonempty, $H_\star + a_{\beta,\gamma} m \ge 0$. For every $u \in \mathcal{A}_{m,H_\star}$,

$$\|\Delta u\|_2 \le \sqrt{H_\star + a_{\beta,\gamma} m} + \kappa\sqrt{m} =: B. \tag{6}$$

On the fixed torus, elliptic regularity gives $\|u\|_{H^2} \le C_\Omega(\sqrt{m} + B)$. The compact embedding $H^2(\mathbb{T}^2) \Subset C^0(\mathbb{T}^2)$ therefore gives a common amplitude bound $\|u\|_\infty \le U$ and subsequential uniform convergence.

3 The fixed-mass nodal theorem

Theorem 3.1 (Uniform nodal coercivity on a fixed Hamiltonian sublevel)**.** *Fix $\Omega, \kappa, \beta, \gamma, m, H_\star, \varepsilon$ as above, with $\mathcal{A}_{m,H_\star} \neq \varnothing$. There exists a number $c_\star = c_\star(\Omega, \kappa, \beta, \gamma, m, H_\star, \varepsilon) > 0$ such that for* every $\alpha \ge 0$ *and every $u \in \mathcal{A}_{m,H_\star}$,*

$$\boxed{\mathcal{Q}_{\varepsilon,\alpha}[u] \ge c_\star \|\Delta u\|_2^2 .} \tag{7}$$

No hypothesis $\inf_\Omega |u| > 0$ is needed. A constructive coefficient is supplied in Theorem 4.1; it depends on the fixed domain spectrum and the resource upper bound.

Proof. From $\alpha \ge 0$ and the common amplitude bound,

$$0 \le |u|^2 + \varepsilon^2 e^{-2\alpha|u|^2} \le U^2 + \varepsilon^2,$$

and therefore

$$\mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{\|u\,\Delta u\|_2^2}{(U^2+\varepsilon^2)^2}. \tag{8}$$

Suppose no positive $c_\star$ works. Then for each integer $j \ge 1$ there are $u_j \in \mathcal{A}_{m,H_\star}$ and $\alpha_j \ge 0$ with $\|\Delta u_j\|_2 > 0$ and

$$\mathcal{Q}_{\varepsilon,\alpha_j}[u_j] < j^{-1} \|\Delta u_j\|_2^2 .$$

The uniform bound (6) implies that the left side tends to zero. By (8), $\|u_j \Delta u_j\|_2 \to 0$. Periodic integration by parts then yields

$$\|\nabla u_j\|_2^2 = -\operatorname{Re}\int_\Omega \overline{u_j}\Delta u_j\,dx \le V^{1/2}\,\|u_j\Delta u_j\|_2 \longrightarrow 0.$$

By the Poincaré inequality, $u_j - \langle u_j\rangle \to 0$ in L^2, where $\langle u\rangle = V^{-1}\int_\Omega u\,dx$. Pass to a subsequence that converges uniformly in C^0, using the H^2 compactness established above. Its limit is spatially constant, and fixed mass forces the modulus of that constant to equal $\sqrt{m/V} > 0$.

Thus, for all sufficiently large j, the *whole* field satisfies $|u_j(x)| \ge \frac{1}{2}\sqrt{m/V}$, for every $x \in \Omega$. Together with the common upper amplitude bound,

$$F_{\varepsilon,\alpha_j}(|u_j|^2) \ge \frac{m}{4V(U^2+\varepsilon^2)^2} =: c_0 > 0 \qquad \text{at every } x \in \Omega.$$

Hence $\mathcal{Q}_{\varepsilon,\alpha_j}[u_j] \ge c_0\,\|\Delta u_j\|_2^2$, contradicting its defining strict inequality for all $j > c_0^{-1}$. □

Remark 3.2 (What fixed mass does and does not imply)**.** *The theorem does* not *prove that each fixed-mass field is node-free. It proves a uniform* integrated *comparison over a compactly controlled class of fields. The positive mass is essential: the earlier high-frequency sequence $N^{-2}\sin(Nx)$ has mass tending to zero and thus does not contradict Theorem 3.1. The Hamiltonian upper bound is essential to prevent the Laplacian energy of arbitrarily fine oscillations from escaping the compact class. The same compactness argument works in dimensions $d \le 3$ on a fixed flat torus; $d = 4$ requires different control because $H^2 \not\hookrightarrow C^0$ there.*

4 An explicit spectral coercivity coefficient

The compactness argument above can be made quantitative on the fixed torus. Let Λ^* be the reciprocal lattice, and use the orthonormal Fourier basis $V^{-1/2}e^{ik\cdot x}$. Let

$$B_+ = 1 + \sqrt{H_\star + a_{\beta,\gamma} m} + \kappa\sqrt{m} > 0, \qquad \mathcal{S}_\Omega = \frac{1}{\sqrt{V}} \left(\sum_{k\in\Lambda^*} \frac{1}{1+|k|^4} \right)^{1/2}, \quad U_+ = \mathcal{S}_\Omega \sqrt{m + B_+^2}.$$

Absolute convergence holds because the torus dimension is two. Parseval and Cauchy–Schwarz show, without invoking an unspecified embedding constant, that every $u \in \mathcal{A}_{m,H_\star}$ satisfies $\|\Delta u\|_2 \le B_+$ and $\|u\|_\infty \le U_+$.

Let $\lambda_1 > 0$ be the first nonzero eigenvalue of $-\Delta$, and set $\tau = \sqrt{m/(32V)}$. Choose a finite radius R containing at least one nonzero reciprocal lattice point, sufficiently large that

$$\mathsf{H}_R B_+ \le \tau, \qquad \mathsf{H}_R = \frac{1}{\sqrt{V}} \left(\sum_{|k|>R} |k|^{-4} \right)^{1/2}.$$

Such an R exists since the Fourier tail converges. Define

$$\mathsf{L}_R = \frac{1}{\sqrt{V}} \left(\sum_{0<|k|\le R} |k|^{-2} \right)^{1/2} > 0, \quad \eta = \min\left\{ \sqrt{\frac{\lambda_1 m}{2}}, \frac{\tau}{\mathsf{L}_R} \right\} > 0.$$

Theorem 4.1 (Explicit fixed-domain coercivity coefficient)**.** *The constant in Theorem 3.1 can be chosen to be*

$$\boxed{c_{\mathrm{exp}} = \frac{1}{(U_+^2+\varepsilon^2)^2} \min\left\{ \frac{m}{8V}, \frac{\eta^4}{VB_+^2} \right\} > 0.} \tag{9}$$

In particular this choice is uniform for all $\alpha \ge 0$*, and makes the dependence on the domain's Fourier spectrum, positive mass, and energy upper bound explicit.*

Proof. Write $g = \|\nabla u\|_2$ and $\bar{u} = V^{-1}\int_\Omega u\,dx$. We distinguish two cases.

If $g \le \eta$, Poincaré and the mass identity give

$$V|\bar{u}|^2 = m - \|u-\bar{u}\|_2^2 \ge m - \lambda_1^{-1} g^2 \ge m/2.$$

Splitting the Fourier series of $u-\bar{u}$ at radius R and applying Cauchy–Schwarz to the gradient and Laplacian terms gives

$$\|u-\bar{u}\|_\infty \le \mathsf{L}_R g + \mathsf{H}_R \|\Delta u\|_2 \le 2\tau = \sqrt{\frac{m}{8V}}.$$

Since $|\bar{u}| \ge \sqrt{m/(2V)}$, the reverse triangle inequality yields $|u(x)| \ge \sqrt{m/(8V)}$ for every x. Consequently

$$\mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{m}{8V(U_+^2+\varepsilon^2)^2} \|\Delta u\|_2^2.$$

If $g > \eta$, periodic integration by parts, Cauchy–Schwarz, and the uniform denominator estimate instead yield

$$\|u\,\Delta u\|_2^2 \ge \frac{g^4}{V} > \frac{\eta^4}{V}, \qquad \mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{\eta^4}{V(U_+^2+\varepsilon^2)^2}.$$

Finally $\|\Delta u\|_2^2 \le B_+^2$, so this implies

$$\mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{\eta^4}{VB_+^2(U_+^2+\varepsilon^2)^2} \|\Delta u\|_2^2.$$

The smaller of the two coefficients proves (9). If $\Delta u = 0$, the assertion is immediate in either case. □

Remark 4.2 (Asymptotic meaning)**.** *This effective constant is generally conservative. It can deteriorate as V grows, $\lambda_1 \to 0$, $H_\star$ grows, or the device geometry changes. Thus the theorem does* not *establish an input-length-uniform physical resource bound. Its merit is that every dependence is now stated as an explicit spectral sum or model parameter.*

5 A detector corollary for the actual Hamiltonian evolution

Let $u(t)$ be a sufficiently smooth periodic solution of (2) on $0 \le t \le T$, conserving positive mass m and Hamiltonian H_0. Let $\chi \in H^4(\Omega; \mathbb{C})$ have zero spatial mean and write

$$Z_\chi(t) = \int_\Omega \overline{\chi}\, u(t)\, dx, \qquad \delta = |Z_\chi(T) - Z_\chi(0)|.$$

Let $\lambda_1 > 0$ be the smallest positive eigenvalue of $-\Delta$, and use the amplitude bound U from the conserved resources. For

$$K_\chi = \frac{\|(\Delta + \kappa)^2 \chi\|_2 + (3\beta U^2 + 5\gamma U^4)\, \|\chi\|_2}{\lambda_1},$$

the periodic equation, the polynomial nonlinear Lipschitz estimate, and the mean-zero Poincaré inequality imply

$$\delta^2 \le K_\chi^2\, T \int_0^T \|\Delta u(t)\|_2^2 \, dt. \tag{10}$$

Corollary 5.1 (Node-tolerant finite-time density-curvature cost)**.** *If $T > 0$, $K_\chi > 0$, $\delta > 0$, and the smooth solution obeys the conserved mass and Hamiltonian hypotheses above, then*

$$\boxed{\int_0^T \mathcal{Q}_{\varepsilon,\alpha}[u(t)]\, dt \ge \frac{c_\star\, \delta^2}{K_\chi^2\, T}.} \tag{11}$$

This remains valid if $u(t, x) = 0$ at some points and times.

Proof. The conserved quantities place every $u(t)$ in the *same* class $\mathcal{A}_{m,H_0}$, so Theorem 3.1 gives one uniform $c_\star > 0$ for all t. Integrate (7) in time and apply (10). □

6 Discrete Lean certificate and exact limitations

On a two-node connected graph with real field values a, b, set $m = a^2 + b^2 > 0$, and use the simplified density weight $w_e(s) = s/(s + e)^2$, $e > 0$. The two graph-Laplacian values are $b - a, a - b$. Since $w_e(a^2) + w_e(b^2) \ge m/(m + e)^2$,

$$\boxed{(a - b)^2[w_e(a^2) + w_e(b^2)] \ge \frac{m}{2(m + e)^2}\, [2(a - b)^2].} \tag{12}$$

The estimate holds even if $a = 0$ or $b = 0$, and the coefficient is strictly positive when $m, e > 0$. The separate Lean module `PVNP/FixedMassNodal.lean` verifies this exact finite result and algebraic detector-to-cost transfer. It does *not* kernel-formalize the Sobolev compactness proof of Theorem 3.1 nor the continuum PDE integration and conservation arguments in Corollary 5.1.

Computational limitation. Neither $c_\star$ nor K_χ is shown to be independent of a growing device domain, input length, resolution, or detector family. The estimates concern a restricted observable of one declared PDE, not all admissible SAT solvers. In particular, no unconditional $P_{\mathrm{WCC}} \ne NP_{\mathrm{WCC}}$ or classical $P \ne NP$ claim follows.

Source provenance. This supplemental note uses the maintained projection-free Hamiltonian equation and follows the independent phase-observability study. The original PVNP trilogy and its published versions are not modified here.